In [2]:
```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score,confusion_matrix
from sklearn.linear_model import LogisticRegression
import seaborn as sns
import matplotlib.pyplot as plt
```

In [3]:
```
df = pd.read_csv("data_airquality.csv")
df.head()
```

```
C:\Users\Dell\AppData\Local\Temp\ipykernel_15196\508662400.py:1: DtypeWarning: Colum
ns (0: stn_code) have mixed types. Specify dtype option on import or set low_memory=
False.
  df = pd.read_csv("data_airquality.csv")
```

Out[3]:

	stn_code	sampling_date	state	location	agency	type	so2	no2
0	150.0	February - M021990	Andhra Pradesh	Hyderabad	NaN	Residential, Rural and other Areas	4.8	17.4
1	151.0	February - M021990	Andhra Pradesh	Hyderabad	NaN	Industrial Area	3.1	7.0
2	152.0	February - M021990	Andhra Pradesh	Hyderabad	NaN	Residential, Rural and other Areas	6.2	28.5
3	150.0	March - M031990	Andhra Pradesh	Hyderabad	NaN	Residential, Rural and other Areas	6.3	14.7
4	151.0	March - M031990	Andhra Pradesh	Hyderabad	NaN	Industrial Area	4.7	7.5

In [4]:
```
#Data cleaning
df=df.drop_duplicates()
df.describe()
```

Out[4]:

	so2	no2	rspm	spm	pm2
count	400436.000000	418846.000000	395033.000000	198160.000000	9314.0000
mean	10.830468	25.823296	108.871711	220.774796	40.7914
std	11.168994	18.497878	74.898637	151.374791	30.8325
min	0.000000	0.000000	0.000000	0.000000	3.0000
25%	5.000000	14.000000	56.000000	111.000000	24.0000
50%	8.000000	22.000000	90.000000	187.000000	32.0000
75%	13.700000	32.200000	142.000000	296.000000	46.0000

max	909.000000	876.000000	6307.033333	3380.000000	504.0000

In [5]:
```
df.isna().sum()
```

Out[5]:
```
stn_code                       144077
sampling_date                       3
state                               0
location                            3
agency                         149466
type                             5357
so2                             34632
no2                             16222
rspm                            40035
spm                            236908
location_monitoring_station     27303
pm2_5                          425754
date                                7
dtype: int64
```

In [6]:
```
percent_missing = df.isnull().sum() * 100 / len(df)
percent_missing.sort_values(ascending=False)
```

Out[6]:
```
pm2_5                          97.859185
spm                            54.453097
agency                         34.354630
stn_code                       33.115973
rspm                            9.202010
so2                             7.960135
location_monitoring_station     6.275571
no2                             3.728613
type                            1.231302
date                            0.001609
sampling_date                   0.000690
location                        0.000690
state                           0.000000
dtype: float64
```

In [7]:
```
df=df.drop(['stn_code', 'agency','sampling_date','location_monitoring_station','pm2
df.head()
```

Out[7]:

	state	location	type	so2	no2	rspm	spm	date
0	Andhra Pradesh	Hyderabad	Residential, Rural and other Areas	4.8	17.4	NaN	NaN	2/1/1990
1	Andhra Pradesh	Hyderabad	Industrial Area	3.1	7.0	NaN	NaN	2/1/1990
2	Andhra Pradesh	Hyderabad	Residential, Rural and other Areas	6.2	28.5	NaN	NaN	2/1/1990
3	Andhra Pradesh	Hyderabad	Residential, Rural and other Areas	6.3	14.7	NaN	NaN	3/1/1990
4	Andhra Pradesh	Hyderabad	Industrial Area	4.7	7.5	NaN	NaN	3/1/1990

In [8]:
```
col_var = ['state', 'location', 'type','date']
col_num = ['so2','no2','rspm','spm']
for col in df.columns:
    if df[col].dtype == 'object' or df[col].dtype == 'string':
        df[col] = df[col].fillna(df[col].mode()[0])
    else:
        df[col] = df[col].fillna(df[col].mean())
```

In [9]:
```
df.isna().sum()
```

Out[9]:
```
state       0
location    0
type        0
so2         0
no2         0
rspm        0
spm         0
date        0
dtype: int64
```

In [10]:
```
#Data integration
subSet1 = df[['state', 'type']]
subSet2 = df[['state','location']]
subSet1.head()
```

Out[10]:

	state	type
0	Andhra Pradesh	Residential, Rural and other Areas
1	Andhra Pradesh	Industrial Area
2	Andhra Pradesh	Residential, Rural and other Areas
3	Andhra Pradesh	Residential, Rural and other Areas
4	Andhra Pradesh	Industrial Area

In [11]:
```
subSet2.head()
```

Out[11]:

	state	location
0	Andhra Pradesh	Hyderabad
1	Andhra Pradesh	Hyderabad
2	Andhra Pradesh	Hyderabad
3	Andhra Pradesh	Hyderabad
4	Andhra Pradesh	Hyderabad

In [12]:
```
concatenated_df = pd.concat([subSet1, subSet2], axis=1)
concatenated_df
```

Out[12]:

	state	type	state	location

0	Andhra Pradesh	Residential, Rural and other Areas	Andhra Pradesh	Hyderabad
1	Andhra Pradesh	Industrial Area	Andhra Pradesh	Hyderabad
2	Andhra Pradesh	Residential, Rural and other Areas	Andhra Pradesh	Hyderabad
3	Andhra Pradesh	Residential, Rural and other Areas	Andhra Pradesh	Hyderabad
4	Andhra Pradesh	Industrial Area	Andhra Pradesh	Hyderabad
...	...	...	...	...
435737	West Bengal	RIRUO	West Bengal	ULUBERIA
435738	West Bengal	RIRUO	West Bengal	ULUBERIA
435739	andaman-and-nicobar-islands	Residential, Rural and other Areas	andaman-and-nicobar-islands	Guwahati
435740	Lakshadweep	Residential, Rural and other Areas	Lakshadweep	Guwahati
435741	Tripura	Residential, Rural and other Areas	Tripura	Guwahati

435068 rows × 4 columns

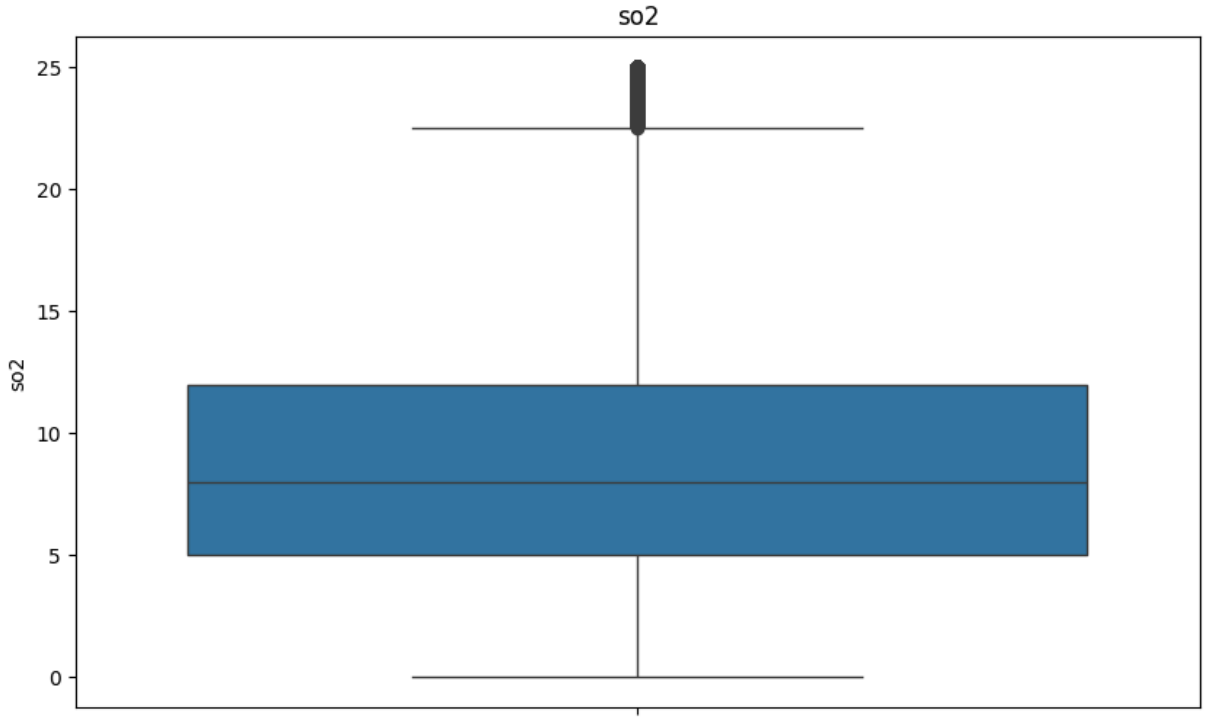
In [13]:
```
#Error correcting
def remove_outliers(column):
    Q1 = column.quantile(0.25)
    Q3 = column.quantile(0.75)
    IQR = Q3 - Q1
    threshold = 1.5 * IQR
    outlier_mask = (column < Q1 - threshold) | (column > Q3 + threshold)
    return column[~outlier_mask]
df.columns
```

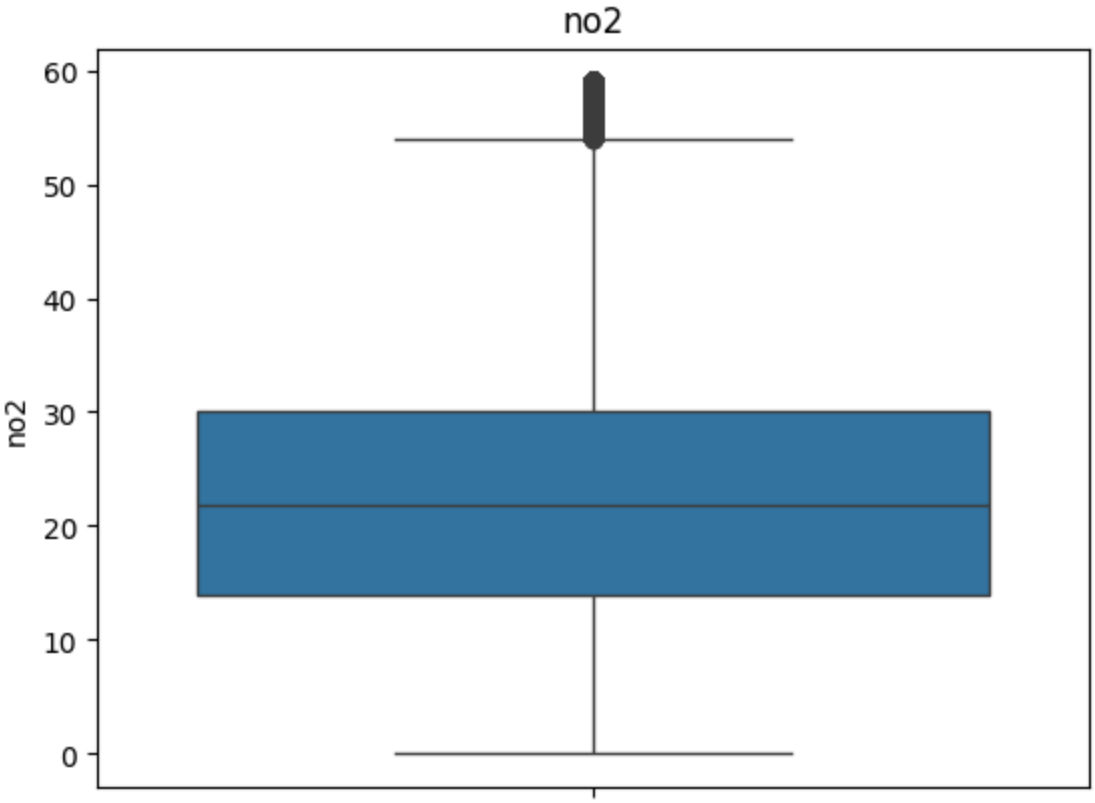
Out[13]:
```
Index(['state', 'location', 'type', 'so2', 'no2', 'rspm', 'spm', 'date'], dtype='s
tr')
```

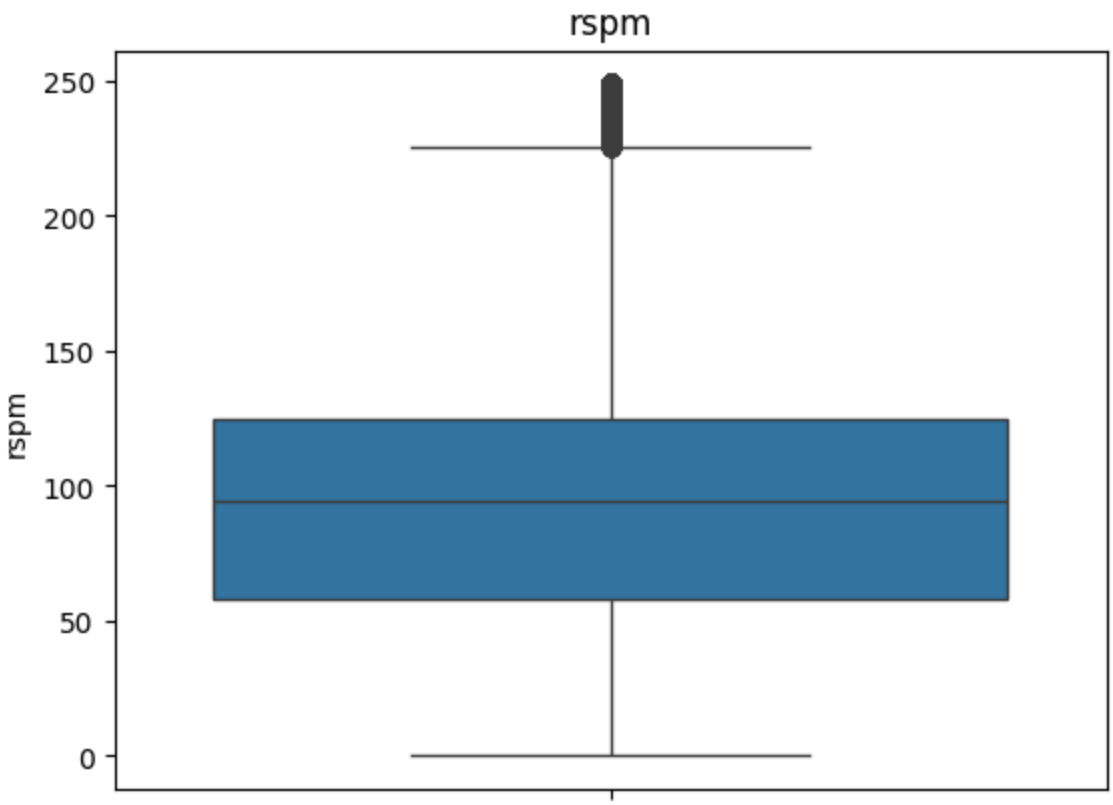
In [14]:
```
#Remove outliers
col_name = ['so2', 'no2', 'rspm', 'spm']
for col in col_name:
    df[col] = remove_outliers(df[col])
```

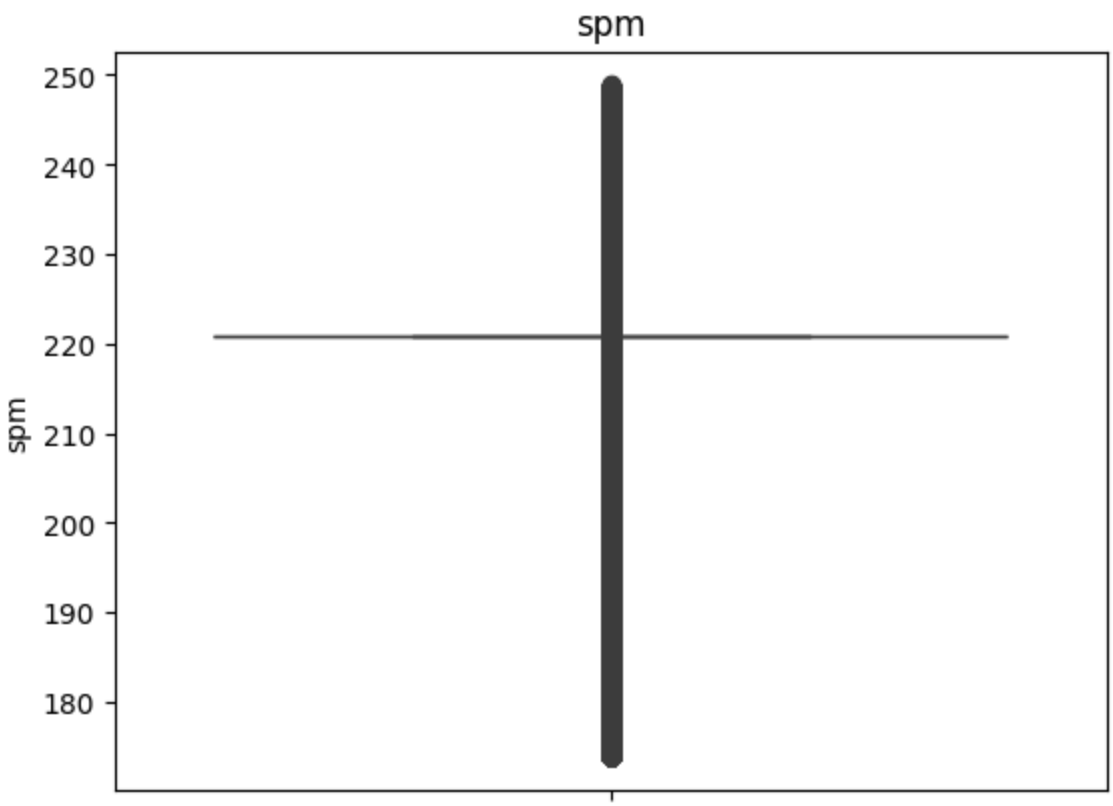
In [15]:
```
plt.figure(figsize=(10, 6))  # Adjust the figure size if needed

for col in col_name:
    sns.boxplot(data=df[col])
    plt.title(col)
    plt.show()
```


so2
so2
25
20
15
10
5
0


no2
no2
60
50
40
30
20
10
0





In [16]:
```
#Data Transform
from sklearn.preprocessing import LabelEncoder

col_label= ['state','location','type']
# Initialize LabelEncoder
```

```
encoder = LabelEncoder()
# Iterate over columns
for col in df.columns:
        # Fit and transform the column
        df[col] = encoder.fit_transform(df[col])
df
```

Out[16]:

	state	location	type	so2	no2	rspm	spm	date
0	0	114	6	446	1489	2030	464	1693
1	0	114	1	197	250	2030	464	1693
2	0	114	6	790	3096	2030	464	1693
3	0	114	6	823	1144	2030	464	2093
4	0	114	1	427	301	2030	464	2093
...	...	...	...	...	...	...	...	...
435737	35	282	3	2888	5307	2534	464	1505
435738	35	282	3	2809	5113	3098	464	1570
435739	36	100	6	1638	2696	2030	464	2256
435740	17	100	6	1638	2696	2030	464	2256
435741	31	100	6	1638	2696	2030	464	2256

435068 rows × 8 columns

In [17]:

```
# splitting data using train test split
df = df.sample(1000, random_state=0)
x = df[['so2','no2','spm','rspm']]
y = df['location']
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=0)

x_train.shape,x_test.shape,y_train.shape,y_test.shape
```

Out[17]:

```
((800, 4), (200, 4), (800,), (200,))
```

In [18]:

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
x_train_scaled = scaler.fit_transform(x_train)
x_test_scaled = scaler.transform(x_test)
```

In [19]:

```
model = LogisticRegression(max_iter=100)
model.fit(x_train_scaled, y_train)
y_pred = model.predict(x_test_scaled)

from sklearn.metrics import accuracy_score
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.075
```

In [ ]: